

Artists' Perspectives on an AI-Powered System for Enhancing Art Exhibitions

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Abstract

We present our preliminary findings from an ongoing study aimed at developing an Artificial Intelligence (AI)-powered system that incorporates artists' input to facilitate meaningful, conversational interactions about artworks, going beyond traditional labels and fostering connections between exhibition visitors and artists. Findings from the formative evaluation of the proposed system with artists ($N=5$) who hold a neutral attitude toward AI suggests that the system has the potential to expand access to and expression of art, act as an agent on behalf of artists, and fit well within their current exhibition practices. These findings may also inform the integration of AI into other artistic practices, including performance art. Based on these findings we discuss the need for educational efforts to expand artists' understanding of AI capabilities beyond generative tools and implications for the future design of the proposed system in terms of preserving artists' authentic voice and supporting multilingual accessibility.

CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; *Empirical studies in interaction design*.

Keywords

Artificial Intelligence, Generative AI, Art Museum, Qualitative Interviews, Formative Evaluation

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1 Introduction

Art exhibitions often provide minimal contextual information about artworks, as curators prioritize aesthetic experience over detailed labeling [10]. This practice, on one hand, limits opportunity for artists to connect with audiences, which is particularly important for emerging artists who often rely on exhibitions to form connection with potential audiences, gaining exposure, and sustaining their careers. On the other hand, it limits visitor engagement [2, 4]. Without adequate insights into contextual information such as creative process, inspirations, or significance of the work, visitors may struggle to form meaningful connections with the artists and their work, diminishing the overall experience. This decline in engagement that has been linked to a global decline in museum attendance, with studies reporting about a 20% drop in recent years [3, 9, 5].

In response to these challenges, art exhibitions are increasingly adopting advanced technologies to provide interactive and immersive experiences. While pre-recorded audio guides to inform visitors about artworks and artists remain common (e.g., [13, 14]), AI-powered mobile application like Smartify [11], used by the Louvre and the Met, allows visitors to scan artworks with their smartphones to instantly access information about the artist, historical context, and related works. Likewise, Useum [16] provides similar features, turning mobile devices into interactive guides that enhance learning and engagement. Some museums have taken this further by leveraging recent advancements in Large Language Models (LLMs). For instance, IBM Brazil developed the “A Voz da Arte” (Voice of Art), which utilizes data from museum archives, news sources, and the internet to answer visitor inquiries about artworks [8]. This system enables users to engage in conversations with historical figures or even with the art itself, offering a novel approach to museum experiences. These initiatives highlight the growing potential of AI in transforming visitor engagement by providing dynamic, personalized interactions during art exhibitions.

However, research suggests that current solutions prioritize object-centered content but lack the depth needed for meaningful engagement [15]. Many rely on online data (e.g., [8]), which can misrepresent artists and their work due to AI hallucination. Additionally, these systems focus primarily on visitor engagement while often excluding artists entirely. As a result, artists have limited opportunities to share their narratives, restricting their ability to express themselves and promote their work. This disconnect prevents these solutions from bridging the gap between audience

curiosity and authentic, artist story, ultimately diminishing both visitor experience and artist exposure.

To address this challenge, we envision an AI-powered system that, unlike existing platforms that rely on scraped data from the internet, integrates input from artists in the form of an essay about their creative processes, inspirations, and the significance of their work, complete with images of their artwork and detailed personal profiles. Leveraging AI capabilities such as image recognition and LLM, the proposed system will engage visitors in meaningful verbal interaction about the artwork and its creator.

In this paper, we describe (i) the design of an AI-powered system centered on artist input, and (ii) findings from a formative study with artists to inform future design of the system.

2 Method

We conducted a formative evaluation of the proposed system through interviews to understand artists' perspectives on AI's role in bridging the gap between audience curiosity and artistic expression as well as its potential to support the promotion of their work.

2.1 Participants

Five artists with expertise in various artistic fields were recruited (See Table 1) through a posters and emails at a university. Participants were compensated with a \$50 gift card for their participation in the study.

Table 1: Participant Demographics

ID	Sex	Education	Specialization
P1	Prefer not to say	MFA	Moving Images, Installation
P2	Female	MFA	Drawing, Painting, Installation
P3	Male	MFA	Painting, Sculpture, Performance Art
P4	Female	BFA	Oil painting, Illustrations
P5	Female	MFA	Sculpture

2.2 Proposed System

As shown in Figure 1, the proposed system consists of a web and mobile application driven by AI backend. The web application enables curators to include direct input from artists in the form of an essay about their creative processes, inspirations, and the significance of their work, along with images of their artworks and detailed personal profiles. This curated content will be used to train a LLM model such as LLaMA (Meta's open-source AI model) within a Retrieval-Augmented Generation (RAG) framework [6]. This approach enhances the accuracy and relevance of inquiries while reducing the risk of hallucinations, a common limitation of LLMs, in contrast to platforms that depend on data scraped from the internet.

During museum visits, users can use the mobile app on their smartphones to scan any artwork they're curious about. The app will recognize the artwork and initiate a dynamic, conversational experience by responding to visitors' questions with insightful responses.

To introduce the proposed system to the study participants, we created a short explainer video¹ demonstrating its functionality

¹<https://youtu.be/Km9ZvAcK5rQ>

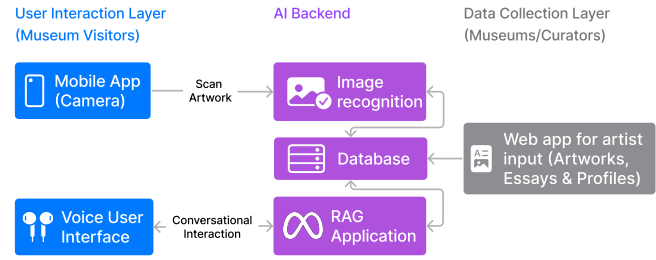


Figure 1: Architecture of the Proposed System.

and usage. We also developed a prototype of a RAG application using Elevenlab's agent tool² to enable participants to experience the proposed system's conversational interactions firsthand. The prototype featured a sample artist and their artwork. For the purposes of this study, features such as image recognition was not included.

2.3 Data Collection & Analysis

During the study session, each participant signed a consent form, completed a demographic questionnaire, and filled out the Attitudes Towards Artificial Intelligence Scale (ATTARI-12) [12] capturing their attitudes toward AI³. ATTARI-12 is a validated questionnaire that includes items such as, Q1: "AI will make this world a better place"; Q2: "I have strong negative emotions about AI"; Q3: "I want to use technologies that rely on AI"; Q4: "AI has more disadvantages than advantages"; Q5: "I look forward to future AI developments"; Q6: "AI offers solutions to many world problems"; Q7: "I prefer technologies that do not feature AI"; Q8: "I am afraid of AI"; Q9: "I would rather choose a technology with AI than one without it"; Q10: "AI creates problems rather than solving them"; Q11: "When I think about AI, I have mostly positive feelings"; and Q12: "I would rather avoid technologies that are based on AI". They watched the explainer video demonstrating the system's functionality in the museum context. They then interacted with the application prototype, followed by audio recorded open-ended interview.

Participants' ATTARI-12 responses were analyzed using R (v. 4.3.2). Items Q2, Q4, Q7, Q8, Q10, and Q12 were reverse-coded prior to analysis. Interview transcripts were analyzed inductively following Braun and Clarke's thematic analysis process [1].

3 Results

Participants in the study generally had a neutral attitude toward AI. Analysis of the interview data revealed that artists perceived the proposed system as a valuable tool with the potential to expand access to and expression of art, act as an agent on behalf of artists, and fit well within their current practices.

3.1 Participants' Attitude towards AI

Figure 2 shows average response scores for each item on the ATTARI-12 questionnaire. The overall mean score across all items was 2.90. Participants reported the highest level of agreement with item Q5

²https://elevenlabs.io/app/talk-to-agent_id=2xK3hMNRfGmD6WEysUej

³P5 was unable to complete this questionnaire due to technical difficulties.

($M=4.25$). Conversely, the lowest average score was observed for item Q9 ($M=2.75$).

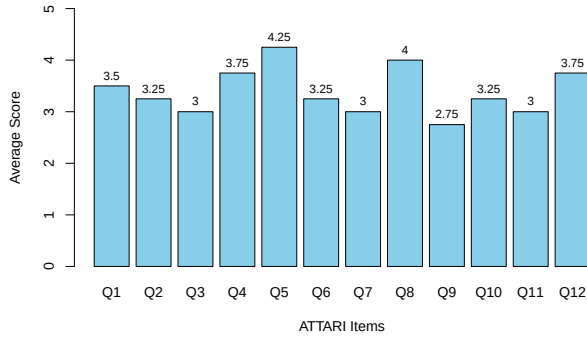


Figure 2: ATTARI-12 Average Score Per Item

3.2 Expanding Access to & Expression of Art

A recurring theme in the interviews was the perception that AI has the potential to make art more accessible to individuals who might not typically engage with it. Participants highlighted that the proposed system's ability to provide contextual information about the artwork being viewed could enhance user engagement and foster learning. P3, for example, explained "There are people who are just like art to them is is not their cup of tea, or it's so foreign to them. So I could see, like, how AI could be helpful in educating them". This sentiment aligns with participants' broader optimism about AI's potential as reflected by higher average score in Q5 of the ATTARI-12.

Artists also mentioned that the proposed system could be useful in helping them articulate their concepts more effectively, especially when they are "stuck in [their] own verbiage on how [they] explain something" [P5]. Participants, however, were concerned that if the AI were to summarize their content or engage with audiences on their behalf, it might not capture the exact tone or language they would use themselves. "I would be kind of concerned on it having more of like a European like viewpoint of how museums look at native art, which I disagree with." [P4]. Despite this concern, they were comfortable with the idea as long as the AI-generated content was based directly on the artists' own input: "if the artist is involved in the whole process, I don't see any harm of that." [P3].

In addition to the proposed system's potential to make art more accessible, artists highlighted several ways it could benefit underrepresented groups. They noted that the proposed system could help individuals overcome language barriers by incorporating translation features, making art more comprehensible across linguistic boundaries. Further, the system's verbal interaction capability would be especially valuable for people who are visually impaired, as it could describe artworks aloud and enhance their experience. This mode of interaction would also support individuals with limited reading proficiency, allowing them to engage more fully with the content.

3.3 AI as Artists' Agent

Another theme of the interviews was the system's potential to represent artists and be their voice, even when they can't be physically present in during exhibitions. Participants believed that the system could foster deeper connections between viewers and creators by verbally offering richer context and personal narratives: "artists may not be able to be there in person, or if they are there in person and they have a crowd around them, it is a way for someone to get the information without speaking directly to the artists..." [P3].

Participants also viewed the system as a meaningful tool for extending their reach with audiences. They emphasized its value in increasing visibility, particularly for underrepresented and emerging artists. As P1 noted, the system could "shine a bigger spotlight" on those still building their presence. Artists wished the system could support a wider range of artists and artworks beyond traditional gallery settings, and create more immersive experiences by integrating with interactive installations and performance arts. P4 explained, "There are other artists doing installations, based on for example sounds...once you explain it to the AI, it can help facilitate things turning on and turning off based on how someone's walking through an installation. And I think that would be really interesting to play with, way beyond the gallery stuff."

Despite this enthusiasm, artists admitted that they are still exploring how best to use AI in their creative practices. When asked how they currently use AI, all participants said they mainly use it for tasks like brainstorming, researching ideas, generating images, and writing proposals. For example, P3 shared, "I use, like, just like the Google Gemini, you know, for brainstorming, kind of, but nothing like when it comes... because I'm a studio artist, you know. Do I use it for that? Not really." Artists also expressed that they would use it more if they knew how to use it effectively in their work. P4 stated: "I would probably use it more if I knew how to use it for sure, so I'm still playing."

3.4 Content Creation as Familiar Practice

The proposed system requires artists to submit materials such as their bio, detailed description of their work, and a digital image of the artwork to enhance the museum visitor experience. Artists generally did not view this process of submitting content to the system as burdensome and reported that the system aligns with the current practice of art exhibition. Participants described having their materials already organized and readily accessible as artists statement about the artwork and their bio. P4 shared, "I have everything saved on my OneDrive, so if anyone needs something, I'm like, here it is. I don't have to keep repeating myself. So it's just a copy and paste for me." For these artists, contributing content to the system was seen as a natural extension of their existing practices.

P2 and P5 however expressed that the requirement to put together detailed written content to train the system could be difficult for artists who are not skilled at writing or describing their work. P5 stated, "Sometimes filling out paperwork, putting words down is very hard, or it could be hard to talk about the piece." P2 added that this requirement might put some artists at a disadvantage, "Some are really great at writing about their work. Some are not. So the person who cannot write about the work can fall behind."

4 Discussion

Findings from the formative evaluation of the proposed system with artists suggests that the proposed system has the potential to expand access to and expression of art, act as an agent on behalf of artists, and fit well within their current exhibition practices. Based on the these findings, we discuss the need for educational efforts to expand artists' understanding of AI capabilities beyond generative tools and implications for the future design of the proposed system in terms of preserving artists' authentic voice and supporting multilingual accessibility. These findings may also inform the integration of AI into other artistic practices, including performance art.

Expanding artists' AI awareness beyond generative tools. Participants in this study had a neutral sentiment towards AI, their higher rating of the ATTARI-12 Q5: "I look forward to future AI developments" and lower rating of Q9: "I would rather choose a technology with AI", suggests that although they are curious and open-minded about AI's potential, they see limited immediate value in its current capabilities for certain art forms including installation and performance art. Such a view towards the current state of AI may be due to a general knowledge gap about AI tools beyond text, image, audio, and video generation, as noted by the study participants. Moreover, the popularity and access to generative AI tools (e.g., ChatGPT, Gemini) may have overshadowed other AI capabilities, such as gesture and motion recognition [7]. As a result, artists have yet to fully explore how AI could support their creative practices including the ones that involves on physical presence and audience engagement. Further research and educational efforts are needed to empower artists by expanding their understanding of AI's wide-ranging possibilities within the creative process. Our ongoing research seeks to address this need by expanding artists' AI knowledge and exploring how AI can effectively support their creative practices.

Preserving artists' authentic voice. Our formative evaluation indicates that the proposed AI-powered system enabling artists' direct input can be a valuable tool for expanding access to and expression of art. However, some artists expressed concerns about a potential mismatch between their own tone and language and how the system might represent them. Current AI tools, such as ElevenLabs⁴ and Synthesia⁵, offer features like voice cloning and custom AI avatar creation, which could help address these concerns.

Support multilingual accessibility. Our findings suggest that while providing information such as an artist's bio and artwork statements aligns well with existing exhibition practices and allows the system to act as an agent for the artist, artists who are less skilled at articulating their work may need support with writing. The system could offer templates and speech-based input to make this process smoother for everyone. Furthermore, to make the system more linguistically inclusive, it could support input and output in various languages on demand, as suggested by participants in this study, using AI translation services such as Google Translate.

⁴<https://elevenlabs.io/voice-cloning>

⁵<https://www.synthesia.io/features/avatars>

5 Limitations

The main limitation of this study is its small sample size ($N = 5$), which means data saturation has not been reached yet. While the findings offer valuable insights into the system's potential to connect artists with their audiences, they do not reflect wider range of artistic practices.

Additionally, the prototype used for evaluation had limited functionality. The image recognition feature was not active, and participants interacted with a predefined example. This may have shaped their perceptions, as they could not assess how the system would represent their unique voice and creative intent.

Despite these limitations, this formative study highlights valuable design considerations for inclusive, artist-centered AI systems. The research is ongoing, and we plan to engage a broader group of artists to better inform system development. We see this work as an important step toward more participatory integration of AI to support and empower artists.

6 Conclusion

This paper presents our preliminary findings from a formative evaluation of an AI powered system designed to connect artists with their audiences. Artists' perception of the system as an agent to expand their reach and make art more accessible to a wider audience suggests that the proposed system holds promise for enhancing museum visits and supporting artist-audience engagement. Through this workshop, we hope to gather more feedback on the ongoing system design, particularly its applicability across different art forms including performance art and contribute to the broader conversation around AI and art.

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